

Hierarchical Access Control

Dr. Ashok Kumar Das

IEEE Senior Member

Professor

Center for Security, Theory and Algorithmic Research
International Institute of Information Technology, Hyderabad

E-mail: *ashok.das@iiit.ac.in*

URL: <http://www.iiit.ac.in/people/faculty/ashokkdas>

Hierarchical Access Control

Overview of Hierarchical Access Control

- Hierarchical access control is a fundamental problem in computer and network systems.
- In a hierarchical access control, a user of higher security level class has the ability to access information items (such as message, data, files, etc.) of other users of lower security classes.
- A user hierarchy consists of a number n of disjoint security classes, say, $SC_1, SC_2, \dots, SC_n$. Let this set be $SC = \{SC_1, SC_2, \dots, SC_n\}$.
- A binary partially ordered relation $\geq$ is defined in SC as $SC_i \geq SC_j$, which means that the security class SC_i has a security clearance higher than or equal to the security class SC_j .

Overview of Hierarchical Access Control

- In addition the relation $\geq$ satisfies the following properties:
 - ▶ **[Reflexive property]** $SC_i \geq SC_i, \forall SC_i \in SC$.
 - ▶ **[Anti-symmetric property]** If $SC_i, SC_j \in SC$ such that $SC_i \geq SC_j$ and $SC_j \geq SC_i$, then $SC_i = SC_j$.
 - ▶ **[Transitive property]** If $SC_i, SC_j, SC_k \in SC$ such that $SC_i \geq SC_j$ and $SC_j \geq SC_k$, then $SC_i \geq SC_k$.
- If $SC_i \geq SC_j$, we call SC_i as the predecessor of SC_j and SC_j as the successor of SC_i . If $SC_i \geq SC_k \geq SC_j$, then SC_k is an intermediate security class. In this case SC_k is the predecessor of SC_j and SC_i is the predecessor of SC_k .
- In a user hierarchy, the encrypted message by a successor security class is only decrypted by that successor class as well as its all predecessor security classes in that hierarchy.

Overview of Hierarchical Access Control

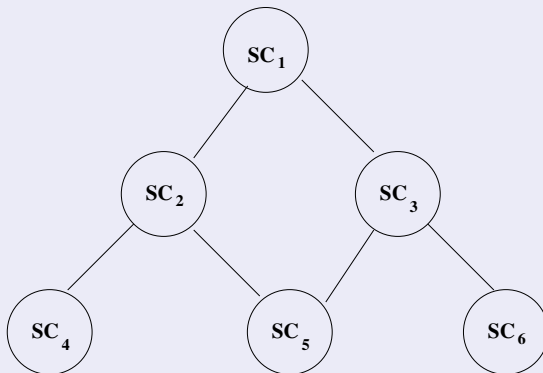


Figure: A small sample of poset in a user hierarchy.

Applications of Hierarchical Access Control

- Military
- Government schools and colleges
- Private corporations
- Computer network systems
- Operating systems
- Database management systems

Chung et al.'s User Hierarchical Access Control Scheme

Reference

- Y. F. Chung, H. H. Lee, F. Lai and T. S. Chen, “Access control in user hierarchy based on elliptic curve cryptosystem”, Information Sciences (Elsevier), vol. 178, no. 1, pp. 230-243, 2008 (2018 SCI Impact Factor: 5.524).

Chung et al.'s User Hierarchical Access Control Scheme

Relationship Building Phase

- CA (central authority) builds a hierarchical structure for controlling access according to the relationships among the nodes in the hierarchy.
- Let $U = \{SC_1, SC_2, \dots, SC_n\}$ be a set of n security classes in the hierarchy. Assume that SC_i is a security class with higher clearance and SC_j a security class with lower clearance, that is, $SC_i \geq SC_j$.
- A legitimate relationship $(SC_i, SC_j) \in R_{i,j}$ between two security classes SC_i and SC_j exists in the hierarchy if SC_i can access SC_j .

Chung et al.'s User Hierarchical Access Control Scheme (Continued...)

Key Generation Phase

CA performs the following steps:

- **Step 1:** Randomly selects a large prime p .
- **Step 2:** Selects an elliptic curve $E_p(a, b)$ defined over Z_p such that the order of $E_p(a, b)$ lies in the interval $[p + 1 - 2\sqrt{p}, p + 1 + 2\sqrt{p}]$.
- **Step 3:** Selects a one-way function $h(\cdot)$ to transform a point into a number and a base point G_j from $E_p(a, b)$ for each security class SC_j $1 \leq j \leq n$.
- **Step 4:** For each security class SC_j ($1 \leq j \leq n$), selects a secret key sk_j and a sub-secret key s_j .
- **Step 5:** For all $\{SC_i | (SC_i, SC_j)\} \in R_{i,j}$, computes the followings:
 $s_i G_j = (x_{j,i}, y_{j,i})$,
 $h(x_{j,i} || y_{j,i})$, where $||$ is a bit concatenation operator.

Chung et al.'s User Hierarchical Access Control Scheme (Continued...)

Key Generation Phase (Continued...)

- **Step 6:** Finally, computes the public polynomial $f_j(x)$ using the values of $h(x_{j,i}||y_{j,i})$ as

$$f_j(x) = \prod_{SC_i \geq SC_j} (x - h(x_{j,i}||y_{j,i})) + sk_j \pmod{p}$$

- **Step 7:** Sends sk_j and s_j to the security class SC_j via a secret channel.
- **Step 8:** Announces $p, h(\cdot), G_j, f_j(x)$ as public.

Chung et al.'s User Hierarchical Access Control Scheme (Continued...)

Key Derivation Phase

In order to compute the secret keys sk_j of all successors, SC_j , the predecessor SC_i , for which the relationships $(SC_i, SC_j) \in R_{i,j}$ between SC_i and SC_j hold, proceeds as follows:

- Step 1: For $\{SC_i | (SC_i, SC_j) \in R_{i,j}\}$, computes the followings:
 $s_i G_j = (x_{j,i}, y_{j,i}),$
 $h(x_{j,i} || y_{j,i}).$
- Step 2: Computes the secret key sk_j using $h(x_{j,i} || y_{j,i})$ as follows:

$$f_j(x) = \prod_{SC_i \geq SC_j} (x - h(x_{j,i} || y_{j,i})) + sk_j \pmod{p},$$
$$f_j(h(x_{j,i} || y_{j,i})) = sk_j \pmod{p}.$$

Chung et al.'s User Hierarchical Access Control Scheme (Continued...)

Key Derivation Phase (Continued...)

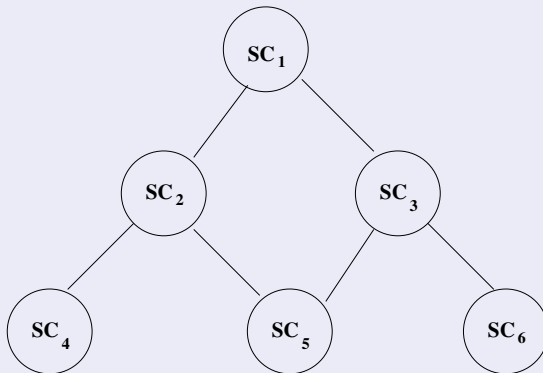


Figure: A small sample of poset in a user hierarchy.

Chung et al.'s User Hierarchical Access Control Scheme (Continued...)

Key Derivation Phase (Continued...)

$$f_j(x) = \prod_{SC_i \geq SC_j} [x - h(x_{j,i} || y_{j,i})] + sk_j \pmod{p},$$

$$SC_1 : f_1(x) = [x - h(x_{1,0} || y_{1,0})] + sk_1 \pmod{p}, \text{ where } s_0 \text{ is given by CA}$$

$$SC_2 : f_2(x) = [x - h(x_{2,1} || y_{2,1})] + sk_2 \pmod{p},$$

$$SC_3 : f_3(x) = [x - h(x_{3,1} || y_{3,1})] + sk_3 \pmod{p},$$

$$SC_4 : f_4(x) = [x - h(x_{4,1} || y_{4,1})][x - h(x_{4,2} || y_{4,2})] + sk_4 \pmod{p},$$

$$SC_5 : f_5(x) = [x - h(x_{5,1} || y_{5,1})][x - h(x_{5,2} || y_{5,2})][x - h(x_{5,3} || y_{5,3})] + sk_5 \pmod{p},$$

$$SC_6 : f_6(x) = [x - h(x_{6,1} || y_{6,1})][x - h(x_{6,3} || y_{6,3})] + sk_6 \pmod{p}$$

Key Derivation Phase (Continued...)

To derive the secret key sk_5 of SC_5 by its predecessor class SC_2 , SC_2 needs to do following:

- Computes $s_2 G_5 = (x_{5,2}, y_{5,2})$ and then $h(x_{5,2} || y_{5,2})$.
- Determines sk_5 using $h(x_{5,2} || y_{5,2})$ from the public polynomial $f_5(x) = [x - h(x_{5,1} || y_{5,1})][x - h(x_{5,2} || y_{5,2})][x - h(x_{5,3} || y_{5,3})] + sk_5 \pmod{p}$ as $sk_5 = f_5(h(x_{5,2} || y_{5,2})) \pmod{p}$.

Inserting New Security Classes Phase

If a new security class SC_k is inserted into the hierarchy such that $SC_i \geq SC_k \geq SC_j$, then the relationships $(SC_i, SC_k) \in R_{i,k}$ for $SC_i \geq SC_k$ and $(SC_k, SC_j) \in R_{k,j}$ for $SC_k \geq SC_j$ need to be updated into the hierarchy. CA needs the following steps to manage the accessing priority of SC_k in the hierarchy.

- Step 1: Updates the partial relationships R that follows when the security class SC_k joins the hierarchy.
- Step 2: Randomly selects the secret key sk_k , the sub-secret key s_k and the base point G_k for the class SC_k .
- Step 3: For all $\{SC_i | (SC_i, SC_k)\} \in R_{i,k}$ that satisfies $SC_i \geq SC_k$ when the new class SC_k is inserted in the hierarchy, computes $s_i G_k = (x_{k,i}, y_{k,i})$, and $h(x_{k,i} || y_{k,i})$.

Chung et al.'s User Hierarchical Access Control Scheme (Continued...)

Inserting New Security Classes Phase (continued...)

- Step 4: Computes the public polynomial $f_k(x)$ as follows:

$$f_k(x) = \prod_{SC_i \geq SC_k} (x - h(x_{k,i} || y_{k,i})) + sk_k \pmod{p}$$

- Step 5: For all $\{SC_i | (SC_i, SC_k)\} \in R_{i,k}$ and $\{SC_k | (SC_k, SC_j)\} \in R_{k,j}$ that satisfy $SC_i \geq SC_k \geq SC_j$ when the new class SC_k is inserted in the hierarchy, computes
 $s_k G_j = (x_{j,k}, y_{j,k})$,
 $s_i G_j = (x_{j,i}, y_{j,i})$,
 $h(x_{j,k} || y_{j,k})$ and $h(x_{j,i} || y_{j,i})$.

Chung et al.'s User Hierarchical Access Control Scheme (Continued...)

Inserting New Security Classes Phase (continued...)

- Step 6: Computes the public polynomial $f'_j(x)$ as follows:

$$f'_j(x) = \prod_{SC_i \geq SC_k \geq SC_j} (x - h(x_{j,i} || y_{j,i}))(x - h(x_{j,k} || y_{j,k})) + sk_j \pmod{p}$$

- Step 7: Replaces $f_j(x)$ with $f'_j(x)$, and sends sk_k and s_k to SC_k via a secure channel, and announces publicly G_k , $f_k(x)$ and $f'_j(x)$.

Chung et al.'s User Hierarchical Access Control Scheme (Continued...)

Inserting New Security Classes Phase (continued...)

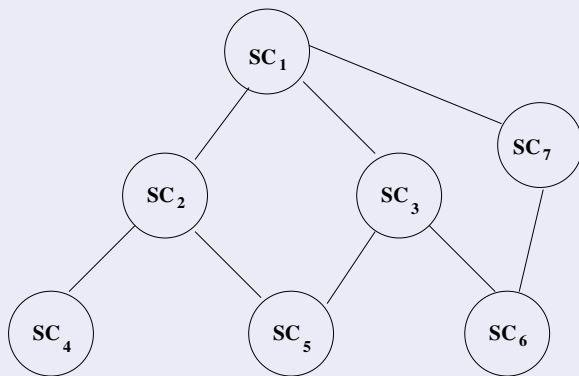


Figure: A small sample of poset in a user hierarchy: when a new security class SC_7 is added into the hierarchy.

Chung et al.'s User Hierarchical Access Control Scheme (Continued...)

Inserting New Security Classes Phase (continued...)

$$SC_1 : f_1(x) = [x - h(x_{1,0} || y_{1,0})] + sk_1 \pmod{p}, \text{ where } s_0 \text{ is given by CA}$$

$$SC_2 : f_2(x) = [x - h(x_{2,1} || y_{2,1})] + sk_2 \pmod{p},$$

$$SC_3 : f_3(x) = [x - h(x_{3,1} || y_{3,1})] + sk_3 \pmod{p},$$

$$SC_4 : f_4(x) = [x - h(x_{4,1} || y_{4,1})][x - h(x_{4,2} || y_{4,2})] + sk_4 \pmod{p},$$

$$SC_5 : f_5(x) = [x - h(x_{5,1} || y_{5,1})][x - h(x_{5,2} || y_{5,2})][x - h(x_{5,3} || y_{5,3})] + sk_5 \pmod{p},$$

$$SC_6 : f'_6(x) = [x - h(x_{6,1} || y_{6,1})][x - h(x_{6,3} || y_{6,3})][x - h(x_{6,7} || y_{6,7})] + sk_6 \pmod{p}$$

$$SC_7 : f_7(x) = [x - h(x_{7,1} || y_{7,1})] + sk_7 \pmod{p}$$

Chung et al.'s User Hierarchical Access Control Scheme (Continued...)

Removing Existing Security Classes Phase

If an existing member SC_k , such that the relationship $SC_i \geq SC_k \geq SC_j$ breaks up, wants to leave from a user hierarchy, then CA not only directly revokes information related to SC_k , but also alters the accessing relationship between the involved ex-predecessor SC_i and ex-successor SC_j of SC_k . In this phase, CA executes the following steps.

- Step 1: Updates the partial relationship R that follows when SC_k is removed.

Chung et al.'s User Hierarchical Access Control Scheme (Continued...)

Removing Existing Security Classes Phase (Continued...)

- Step 2: For all $\{SC_k|(SC_k, SC_j)\} \in R_{k,j}$ does the followings:
 - ▶ Step 2.1: Renews the secret key sk_j as sk'_j and the base point G_j as G'_j of SC_j .
 - ▶ Step 2.2: For all $\{SC_i|(SC_i, SC_j)\} \in R_{i,j}$ does the followings:
 - ★ Step 2.2.1: Renews $\{SC_i|(SC_i, SC_j)\} \in R_{i,j}$ after removing SC_k .
 - ★ Step 2.2.2: Computes $s_i G'_j = (x_{j,i}, y_{j,i})$.
 - ★ Step 2.2.3: Computes $h(x_{j,i}, y_{j,i})$.
 - ★ Step 2.2.4: Computes the public polynomial $f'_j(x)$ as
$$f'_j(x) = \prod_{SC_i \geq SC_j} (x - h(x_{j,i}||y_{j,i})) + sk'_j \pmod{p}$$
 - ★ Step 2.2.5: Replaces $f_j(x)$ with $f'_j(x)$.
- Step 3: Sends sk'_j to SC_j via a secret channel and announces G'_j and $f'_j(x)$ as public.

Removing Existing Security Classes Phase (continued...)

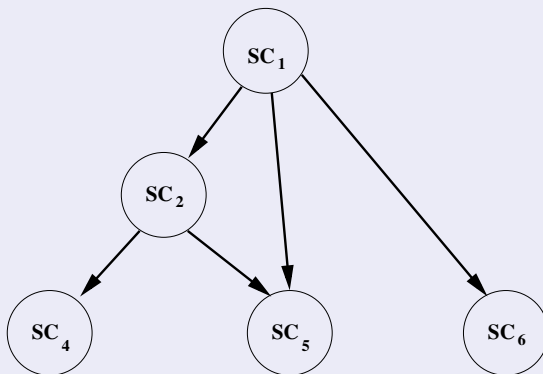


Figure: A small sample of poset in a user hierarchy: when an existing security class SC_3 is removed into the hierarchy.

Removing Existing Security Classes Phase (continued...)

- Before deleting SC_3 , $f_5(x)$ and $f_6(x)$ are formed as

$$f_5(x) = [x - h(x_{5,1}||y_{5,1})][x - h(x_{5,2}||y_{5,2})][x - h(x_{5,3}||y_{5,3})] + sk_5 \pmod{p}$$

$$f_6(x) = [x - h(x_{6,1}||y_{6,1})][x - h(x_{6,3}||y_{6,3})] + sk_6 \pmod{p}.$$

- After deleting SC_3 , $f'_5(x)$ and $f'_6(x)$ are formed as

$$f'_5(x) = [x - h(x_{5,1}||y_{5,1})][x - h(x_{5,2}||y_{5,2})] + sk'_5 \pmod{p}$$

$$f'_6(x) = [x - h(x_{6,1}||y_{6,1})] + sk'_6 \pmod{p}.$$

Creating New Relationships

- Suppose we want to create a new relationship between SC_5 and SC_6 in the hierarchy (Figure 1) such that $SC_2 \geq SC_5 \geq SC_6$.

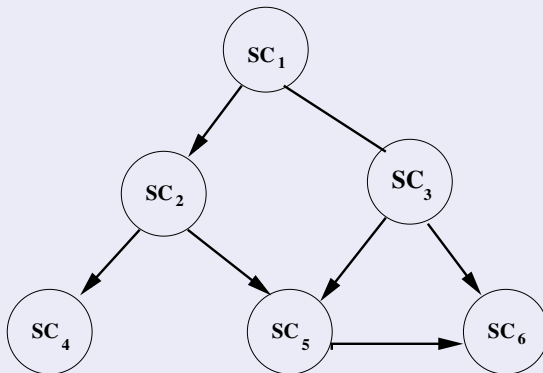


Figure: The consequent poset after creating $SC_5 \geq SC_6$ in Figure 1.

Creating New Relationships

- Before creating the relationship $SC_2 \geq SC_5 \geq SC_6$, $f_6(x)$ is formed as follows:

$$f_6(x) = [x - h(x_{6,1} || y_{6,1})][x - h(x_{6,3} || y_{6,3})] + sk_6 \pmod{p}.$$

- After creating the relationship $SC_2 \geq SC_5 \geq SC_6$, updated public polynomial $f'_6(x)$ is formed as follows:

$$f'_6(x) = [x - h(x_{6,1} || y_{6,1})][x - h(x_{6,3} || y_{6,3})] \\ [x - h(x_{6,2} || y_{6,2})][x - h(x_{6,5} || y_{6,5})] + sk_6 \pmod{p}.$$

Revoking Existing Relationships

- Suppose we want to revoke the existing relationship $\{SC_2 | (SC_2, SC_5) \in R_{2,5}\}$ in the following figure such that $\{SC_2 | (SC_2, SC_5) \notin R_{2,5}\}$

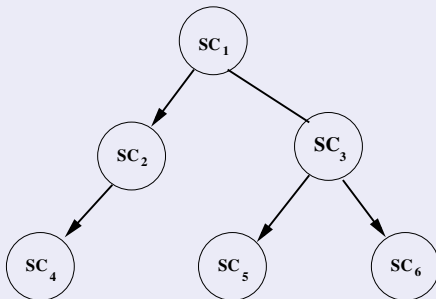


Figure: The consequent poset after revoking $SC_2 \geq SC_5$ in Figure 1.

Revoking Existing Relationships

- Before revoking $\{SC_2 | (SC_2, SC_5) \in R_{2,5}\}$, $f_5(x)$ is formed as follows:

$$f_5(x) = [x - h(x_{5,1} || y_{5,1})][x - h(x_{5,2} || y_{5,2})][x - h(x_{5,3} || y_{5,3})] + sk_5 \pmod{p}.$$

- After revoking $\{SC_2 | (SC_2, SC_5) \in R_{2,5}\}$, $f_5(x)$ is replaced with the updated $f'_5(x)$ as follows:

$$f'_5(x) = [x - h(x_{5,1} || y_{5,1})][x - h(x_{5,3} || y_{5,3})] + sk'_5 \pmod{p}.$$

after renewing the secret key sk'_5 in place of sk_5 .

Changing Secret Keys

- A secret key must be changeable to maximize security.
- To change a secret key sk_j to sk'_j , CA must replace the base point G_j with G'_j and the public polynomial $f_j(x)$ with $f'_j(x)$ as follows.
 - ▶ Step 1: Replace the secret key sk_j with sk'_j and the base point G_j with G'_j .
 - ▶ Step 2: For all $\{SC_i | (SC_i, SC_j)\} \in R_{i,j}$:
 - ★ Step 2.1: Determine $s_i G'_j = (x_{j,i}, y_{j,i})$
 - ★ Step 2.2: Determine $h(x_{j,i} || y_{j,i})$, where $||$ is a bit concatenation operator

Changing Secret Keys (Continued...)

- Step 3: Determine the public polynomial $f'_j(x)$ as follows

$$f'_j(x) = \prod_{SC_i \geq SC_j} [x - h(x_{j,i} || y_{j,i})] + sk'_j \pmod{p}$$

- Step 4: Replace $f_j(x)$ with $f'_j(x)$
- Step 5: Send sk'_j to SC_j via a secret channel, and announce G'_j and $f'_j(x)$

Cryptanalysis and Improvement of Chung et al.'s Scheme

- **Ashok Kumar Das, Nayan Ranjan Paul, and Laxminath Tripathy. “Cryptanalysis and improvement of an access control in user hierarchy based on elliptic curve cryptosystem,” in *Information Sciences (Elsevier)*, Vol. 209, No. C, pp. 80 - 92, 2012, doi: <http://dx.doi.org/10.1016/j.ins.2012.04.036>. (2018 SCI Impact Factor: 5.524)**

Cryptanalysis and Improvement of Chung et al.'s Scheme

Root finding of a polynomial in finite field $GF(q)$

- Suppose we are given a polynomial $f(x) \in GF(q)[x]$ of degree n ($\deg.f = n$), in $GF(q)$ and we want to find all the roots $a \in GF(q)$ of $f(x) = 0$. Let q be odd.
- M. Ben-or showed that the expected number of operations used by his proposed algorithm for finding all roots of $f(x) = 0$ is $O(n \cdot \log n \cdot L(n) \cdot \log q)$, where $L(n) = \log n \cdot \log \log n$. In other words, the expected running time of the algorithm is polynomial in n .

Reference: M. Ben-Or, Probabilistic algorithms in finite fields, in: Proceedings of 22nd Annual Symposium on Foundations of Computer Science (IEEE FOCS'81), 1981, pp. 394-398.

Cryptanalysis and Improvement of Chung et al.'s Scheme

Root finding of a polynomial in finite field $GF(q)$

Algorithm: Finding roots of $f_1(x) \in GF(q)[x] : ROOTS(f_1(x))$

if ($\deg.f_1(x) = 1$, i.e., $f_1(x) = (x - a)$) **then**

return a ;

end if

repeat

 Choose $\delta \in GF(q)$ randomly;

 Compute $f_2(x) = \gcd[f_1(x), (x + \delta)^{(q-1)/2} - 1]$;

until ($0 < \deg.f_2 < \deg.f_1$)

return $ROOTS(f_2(x)) \cup ROOTS(f_1(x)/f_2(x))$;

Cryptanalysis and Improvement of Chung et al.'s Scheme

Factorization of a polynomial in finite field $GF(q)$

- Let we want to factor a polynomial $f(x) \in GF(q)[x]$ of degree n into its irreducible factors. Assume that $f(x)$ has no repeated factors and we want to factor $f(x)$ as $f(x) = g_1(x).g_2(x) \dots g_n(x)$, where $g_d(x)$ is the product of irreducible factors of $f(x)$ of degree d .
- M. Ben-or showed that the expected number of operations to factor an n -degree polynomial is $O(n^2.L(n). \log q)$. That is, the expected running time of the algorithm is also polynomial in n .

Reference: M. Ben-Or, Probabilistic algorithms in finite fields, in: Proceedings of 22nd Annual Symposium on Foundations of Computer Science (IEEE FOCS'81), 1981, pp. 394-398.

Cryptanalysis and Improvement of Chung et al.'s Scheme

Factorization of a polynomial in finite field $GF(q)$

Algorithm: Factorization

$$r_0(x) = x;$$

$$f_0(x) = f(x);$$

for $d = 1 \rightarrow n$ **do**

$$r_d(x) = (r_{d-1}(x))^q \pmod{f_{d-1}(x)}, \deg.r_d < \deg.f_{d-1};$$

$$g_d(x) = \gcd(f_{d-1}(x), r_d(x) - x);$$

$$f_d(x) = f_{d-1}(x)/g_d(x);$$

end for

return $g_1(x), g_2(x), \dots, g_n(x);$

Cryptanalysis and Improvement of Chung et al.'s Scheme

The exterior root finding attack on Chung et al.'s scheme

- Consider the case when a security class SC_k is inserted in the user hierarchy with the relationship $SC_i \geq SC_k \geq SC_j$.
- Thus, when a new security class SC_k is added as a predecessor of SC_j , CA updates the public polynomial of SC_j by replacing $f_j(x)$ by $f'_j(x)$.
- However, for those predecessors, which remain as predecessors of SC_j in $f'_j(x)$, their secrets are still at the same positions of $f'_j(x)$.
- Now, knowing the public polynomial $f_j(x)$ of SC_j before adding the security class SC_k and the public polynomial $f'_j(x)$ of SC_j after adding the security class SC_k until the secret key sk_j of SC_j has been changed by CA, an attacker can generate a polynomial by taking the difference of $f_j(x)$ and $f'_j(x)$.
- Let this difference polynomial be denoted by $\phi(x) = f_j(x) - f'_j(x)$.

Cryptanalysis and Improvement of Chung et al.'s Scheme

Exterior root finding attack on Chung et al.'s scheme

- It is noted that

$$\begin{aligned}\phi(x) &= f_j(x) - f'_j(x) \\ &= \left(\prod_{SC_i > SC_j} [x - h(x_{j,i} || y_{j,i})] + sk_j \pmod{p} \right) - \\ &\quad \left(\prod_{SC_i > SC_k > SC_j} [x - h(x_{j,i} || y_{j,i})][x - h(x_{j,k} || y_{j,k})] \right. \\ &\quad \left. + sk_j \pmod{p} \right) \\ &= \prod_{SC_i > SC_j} [x - h(x_{j,i} || y_{j,i})] - \\ &\quad \prod_{SC_i > SC_k > SC_j} [x - h(x_{j,i} || y_{j,i})][x - h(x_{j,k} || y_{j,k})] \pmod{p}\end{aligned}$$

Cryptanalysis and Improvement of Chung et al.'s Scheme

Exterior root finding attack on Chung et al.'s scheme

- Further, we observe that the constructed polynomial $\phi(x)$ has common factors $(x - h(x_{j,i}||y_{j,i}))$. Then the attacker finds the roots of the equation $\phi(x) = f_j(x) - f'_j(x) = 0$ in a polynomial time using M. Ben-or's method.
- With the knowledge of the roots, the attacker can easily derive the secret key sk_j of the security class SC_j .
- The attacker, who is not a user in hierarchy of security classes, first obtains the roots $h(x_{j,i}||y_{j,i})$ and then computes the secret key sk_j of SC_j as $sk_j = f_j(h(x_{j,i}||y_{j,i})) = f'_j(h(x_{j,i}||y_{j,i})) \pmod{p}$.

Cryptanalysis and Improvement of Chung et al.'s Scheme

Example [Exterior root finding attack on Chung et al.'s scheme]

- In Figure 1, the user hierarchy consists of six security classes, denoted by $U = \{SC_1, SC_2, SC_3, SC_4, SC_5, SC_6\}$.

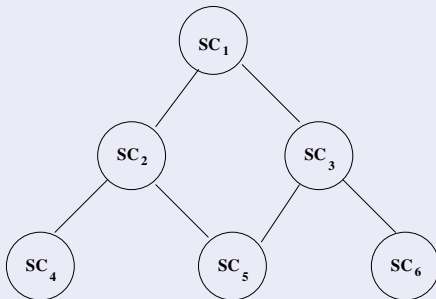


Figure: A small sample of poset in a user hierarchy.

Cryptanalysis and Improvement of Chung et al.'s Scheme

Example [Exterior root finding attack on Chung et al.'s scheme]

- CA computes the public elliptic curve polynomial $f_j(x)$ for each SC_j . Each SC_i derives the secret keys of its successors SC_j :

$$SC_1 : f_1(x) = [x - h(x_{1,0} || y_{1,0})] + sk_1 \pmod{p}, \text{ where } s_0 \text{ is given by CA}$$

$$SC_2 : f_2(x) = [x - h(x_{2,1} || y_{2,1})] + sk_2 \pmod{p},$$

$$SC_3 : f_3(x) = [x - h(x_{3,1} || y_{3,1})] + sk_3 \pmod{p},$$

$$SC_4 : f_4(x) = [x - h(x_{4,1} || y_{4,1})][x - h(x_{4,2} || y_{4,2})] + sk_4 \pmod{p},$$

$$SC_5 : f_5(x) = [x - h(x_{5,1} || y_{5,1})][x - h(x_{5,2} || y_{5,2})][x - h(x_{5,3} || y_{5,3})] + sk_5 \pmod{p},$$

$$SC_6 : f_6(x) = [x - h(x_{6,1} || y_{6,1})][x - h(x_{6,3} || y_{6,3})] + sk_6 \pmod{p}$$

Cryptanalysis and Improvement of Chung et al.'s Scheme

Example [Exterior root finding attack on Chung et al.'s scheme]

- Suppose a new security class SC_7 is inserted into the existing user hierarchy shown in Figure 1, with the relationship $SC_1 \geq SC_7 \geq SC_6$. The resulting user hierarchy is shown below.

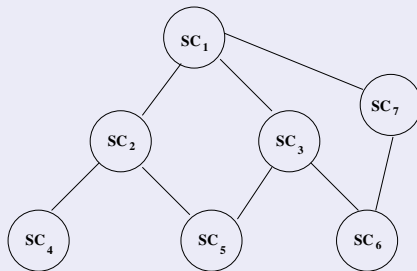


Figure: A small sample of poset in a user hierarchy: when a new security class SC_7 is added into the hierarchy.

Cryptanalysis and Improvement of Chung et al.'s Scheme

Example [Exterior root finding attack on Chung et al.'s scheme]

- Due to insertion of SC_7 , CA needs to select randomly sk_7 , s_7 and G_7 . Since SC_7 is a successor of SC_1 and a predecessor of SC_6 , CA constructs the public polynomial $f_7(x)$ and replaces the public polynomial $f_6(x)$ with $f'_6(x)$.
- Before joining the security class SC_7 into the hierarchy, the public elliptic curve polynomial for security class SC_6 was

$$f_6(x) = [x - h(x_{6,1}||y_{6,1})][x - h(x_{6,3}||y_{6,3})] + sk_6 \pmod{p} \quad (1)$$

- After joining the security class SC_7 , the public polynomial $f'_6(x)$ for SC_6 and $f_7(x)$ for SC_7 are formed as follows:

$$\begin{aligned} f'_6(x) &= [x - h(x_{6,1}||y_{6,1})][x - h(x_{6,3}||y_{6,3})] \\ &\quad [x - h(x_{6,7}||y_{6,7})] + sk_6 \pmod{p} \end{aligned} \quad (2)$$

$$f_7(x) = [x - h(x_{7,1}||y_{7,1})] + sk_7 \pmod{p} \quad (3)$$

Cryptanalysis and Improvement of Chung et al.'s Scheme

Example [Exterior root finding attack on Chung et al.'s scheme]

- Knowing the public polynomials $f_6(x)$ and $f'_6(x)$ in Equations (4) and (5), an attacker finds the roots of the equation

$$\begin{aligned}\phi(x) &= f_6(x) - f'_6(x) = 0 \\ \Rightarrow & [x - h(x_{6,1}||y_{6,1})][x - h(x_{6,3}||y_{6,3})] \\ & [1 - (x - h(x_{6,7}||y_{6,7}))] = 0 \pmod{p}\end{aligned}\tag{4}$$

- The attacker obtains the roots as $x = h(x_{6,1}||y_{6,1})$, $h(x_{6,3}||y_{6,3})$ and $1 + h(x_{6,7}||y_{6,7})$. Out of these roots, $h(x_{6,1}||y_{6,1})$ and $h(x_{6,3}||y_{6,3})$ satisfy both Equations (4) and (5).

Cryptanalysis and Improvement of Chung et al.'s Scheme

Example [Exterior root finding attack on Chung et al.'s scheme]

- Thus, knowing these values, the attacker easily computes the secret key sk_6 of security class SC_6 as

$$\begin{aligned} sk_6 &= f_6(h(x_{6,1} || y_{6,1})) \pmod{p} \\ &= f'_6(h(x_{6,1} || y_{6,1})) \pmod{p} \\ &= f_6(h(x_{6,3} || y_{6,3})) \pmod{p} \\ &= f'_6(h(x_{6,3} || y_{6,3})) \pmod{p}. \end{aligned}$$

Cryptanalysis and Improvement of Chung et al.'s Scheme

Improvement of Chung et al.'s Scheme

- Improvement over Chung et al.'s scheme is provided in the inserting new security classes phase in order to withstand the exterior root finding attack on Chung et al.'s scheme.
- For more details, see the paper:
Ashok Kumar Das, Nayan Ranjan Paul, and Laxminath Tripathy. "Cryptanalysis and improvement of an access control in user hierarchy based on elliptic curve cryptosystem," in *Information Sciences (Elsevier)*, Vol. 209, No. C, pp. 80 - 92, 2012, doi: <http://dx.doi.org/10.1016/j.ins.2012.04.036>. (2018 SCI Impact Factor: 5.524)

Cryptanalysis and Improvement of Chung et al.'s Scheme

Improvement of Chung et al.'s Scheme

Inserting new security classes: If a new security class SC_k is inserted into the hierarchy such that $SC_i \geq SC_k \geq SC_j$, then the relationships $(SC_i, SC_k) \in R_{i,k}$ for $SC_i \geq SC_k$ and $(SC_k, SC_j) \in R_{k,j}$ for $SC_k \geq SC_j$ need to be updated into the hierarchy. In this phase, CA renews the secret keys sk_j of successors SC_j of the newly added security class SC_k . Moreover, CA must change the public base points G_j by G'_j and the public elliptic curve polynomials $f_j(x)$ with $f'_j(x)$ of SC_j . CA needs the following steps:

- Step 1: Updates the partial relationships R that follow when the security class SC_k joins the hierarchy.
- Step 2: Randomly selects the secret key sk_k , the sub-secret key s_k and the base point G_k for the class SC_k .
- Step 3: For all $\{SC_i | (SC_i, SC_k) \in R_{i,k}\}$ that satisfies $SC_i \geq SC_k$ when the new class SC_k is inserted in the hierarchy, computes $s_i G_k = (x_{k,i}, y_{k,i})$, and $h(x_{k,i} || y_{k,i})$.

Improvement of Chung et al.'s Scheme

- Step 4: Computes the public polynomial $f_k(x)$ as follows:

$$f_k(x) = \prod_{SC_i > SC_k} (x - h(x_{k,i} || y_{k,i})) + sk_k \pmod{p}$$

- Step 5: For all $\{SC_i | (SC_i, SC_k) \in R_{i,k}\}$ and $\{SC_j | (SC_k, SC_j) \in R_{k,j}\}$ that satisfy $SC_i \geq SC_k \geq SC_j$ when the new class SC_k is inserted in the hierarchy:

Replaces the secret key sk_j with sk'_j and the base point G_j with G'_j of the successor security class SC_j of SC_k .

Computes $s_k G'_j = (x'_{j,k}, y'_{j,k})$.

Computes $s_i G'_j = (x'_{j,i}, y'_{j,i})$.

Computes $h(x'_{j,k} || y'_{j,k})$ and $h(x'_{j,i} || y'_{j,i})$ using the one-way function $h(\cdot)$.

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Improvement of Chung et al.'s Scheme

- Step 6: Computes the public polynomial $f'_j(x)$ as follows:

$$f'_j(x) = \prod_{SC_i > SC_k > SC_j} (x - h(x'_{j,i} || y'_{j,i}))(x - h(x'_{j,k} || y'_{j,k})) + sk'_j \pmod{p}$$

- Step 7: Replaces $f_j(x)$ with $f'_j(x)$, and sends sk'_j to SC_j via a secure channel, and announces publicly G'_j and $f'_j(x)$.
- Step 8: Sends sk_k and s_k to SC_k via a secure channel, and announces publicly G_k and $f_k(x)$.

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Cryptanalysis and Improvement of Chung et al.'s Scheme

Key derivation: For the relationship $(SC_i, SC_j) \in R_{i,j}$ between two security classes SC_i and SC_j , if the predecessor SC_i wants to compute the updated secret key sk'_j of its successor SC_j , then SC_i needs to proceed the following steps:

- Step 1: For $\{SC_i | (SC_i, SC_j) \in R_{i,j}\}$, computes $s_i G'_j = (x'_{j,i}, y'_{j,i})$ and $h(x'_{j,i} || y'_{j,i})$.
- Step 2: Computes the secret key sk'_j using the computed hash value $h(x'_{j,i} || y'_{j,i})$ as follows:
 SC_i has the updated public elliptic curve polynomial for SC_j as

$$f'_j(x) = \prod_{SC_i > SC_k > SC_j} (x - h(x'_{j,i} || y'_{j,i}))(x - h(x'_{j,k} || y'_{j,k})) + sk'_j \pmod{p}.$$

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- Step 3: Computes

$$sk'_j = f'_j(h(x'_{j,i} || y'_{j,i})) \pmod{p}.$$

Important References

- N. Koblitz, “Elliptic Curve Cryptosystems”, Mathematics of Computation, vol. 48, pp. 203-209, 1987.
- V. Miller, “Uses of elliptic curves in cryptography”, in Proceedings of Advances in Cryptology (CRYPTO’85), LNCS 218, pages 417-426, 1986, Springer-Verlag.
- Y. F. Chung, H. H. Lee, F. Lai and T. S. Chen, “Access control in user hierarchy based on elliptic curve cryptosystem”, Information Sciences (Elsevier), vol. 178, no. 1, pp. 230-243, 2008. (2018 SCI Impact Factor: 5.524)
- Ashok Kumar Das, Nayan Ranjan Paul, and L. Tripathy. “Cryptanalysis and improvement of an access control in user hierarchy based on elliptic curve cryptosystem,” in Information Sciences (Elsevier), Vol. 209, No. C, pp. 80 - 92, 2012. (2018 SCI Impact Factor: 5.524)
- Vanga Odelu, Ashok Kumar Das, and A. Goswami. “A secure effective key management scheme for dynamic access control in a large leaf class hierarchy,” in Information Sciences (Elsevier), Vol. 269, No. C, pp. 270-285, 2014. (2018 SCI Impact Factor: 5.524)

- S.G. Akl, P.D. Taylor, Cryptographic solution to a problem of access control in a hierarchy, ACM Transactions on Computer Systems 1 (3) (1983) 239-248.
- S. Mackinnon, P. Taylor, H. Meijer, S. Akl, An optimal algorithm for assigning cryptographic keys to control access in a hierarchy, IEEE Transactions on Computers 34 (9) (1985) 797-802.
- Y.-L. Lin, C.-L. Hsu, Secure key management scheme for dynamic hierarchical access control based on ECC, Journal of Systems and Software 84 (4) (2011) 679-685.
- J.-W. Lo, M.-S. Hwang, C.-H. Liu, An efficient key assignment scheme for access control in a large leaf class hierarchy, Information Sciences 181 (2011) 917-925.
- M. Nikooghadam, A. Zakerolhosseini, Secure Communication of Medical Information using Mobile Agents, Journal of Medical Systems 36 (6) (2012) 3839-3850.

- M. Nikooghadam, A. Zakerolhosseini, M.E. Moghaddam, Efficient utilization of elliptic curve cryptosystem for hierarchical access control, Journal of Systems and Software 83 (10) (2010) 1917-1929.
- R.S. Sandhu, Cryptographic implementation of a tree hierarchy for access control, Information Processing Letters 27 (2) (1988) 95-98.
- Vanga Odelu, Ashok Kumar Das, and Adrijit Goswami. "An Effective and Secure Key-Management Scheme for Hierarchical Access Control in E-Medicine System," in Journal of Medical Systems (Springer), Vol. 37, No. 2, pp. 1 - 18, 2013. (2016 SCI Impact Factor: 2.456)

Thank you!